ECOSYSTEM SURVIVAL

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Introduction.

The Ecosystem Survival software is an interactive grid-based system designed to simulate wildlife management within a controlled environment. Built on an 8 x 8 grid, the application models the behavior and survival conditions of four forest species: Rabbit, Fox, Turtle, and Deer. The system integrates classical algorithmic strategies with structured data management to provide a dynamic and decision-driven simulation, where each action directly impacts ecosystem stability and species survival.

Goal

The objective is to combine algorithm design with efficient data structures to manage the ecosystem. An AVL tree ensures O(log n) operations for population control, providing accuracy and performance.

System Architecture

The system will use a hybrid architecture with a C++ engine and a Python interface. C++ manages the AVL tree and linked list, while Python handles algorithms and interaction. Both communicate through JSON files (input.json and state.json).